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1. Prove that the linear congruence $ax \equiv b \pmod{n}$ has a solution if and only if $\gcd(a, n)$ divides b .

Solution: Necessity: If $ax \equiv b \pmod{n}$ has a solution, then $ax = b + kn$ for some integer x and k , or equivalently, $ax - kn = b$. Since the $d = \gcd(a, n)$ divides both a and n , it must divide b for this equation to hold.

Sufficiency: If $d = \gcd(a, n)$ divides b , then by Bézout’s identity, there exist integers x_0 and y_0 such that:

$$ax_0 + ny_0 = d.$$

Multiplying this equation by $\frac{b}{d}$ gives a particular solution to the congruence.

2. For a positive integer n and a prime number p , let $v_p(n)$ denote the exponent of p which appears in the prime factorization of n . Note that by uniqueness in the Fundamental Theorem of Arithmetic this function is well-defined.

Let a and b be positive integers. Prove or disprove that $a|b$ if and only if $v_p(a) \leq v_p(b)$ for every prime number p such that $p|a$.

Solution: First suppose that $a|b$ and let p be a prime number such that $p|a$. Then we have that $p^{v_p(a)}|a$ and $a|b$ so $p^{v_p(a)}|b$ and so by definition of v_p we get that $p^{v_p(a)}|p^{v_p(b)}|b$ and so $v_p(b) \geq v_p(a)$. Suppose instead that $v_p(a) \leq v_p(b)$ for every p such that $p|a$. Let $\{p_1, \dots, p_k\}$ denote the prime numbers appearing in the unique factorization of a so that $a = p_1^{v_{p_1}(a)} \dots p_k^{v_{p_k}(a)}$. Since $v_{p_i}(a) \leq v_{p_i}(b)$ for every p_i we have that $a|p_1^{v_{p_1}(b)} \dots p_k^{v_{p_k}(b)}$, furthermore, each $p_i^{v_{p_i}(b)}$ is a factor of b by definition of v_{p_i} and $\gcd(p_i^{v_{p_i}(b)}, p_k^{v_{p_k}(b)}) = 1$ for each $i \neq k$ and so $p_1^{v_{p_1}(b)} \dots p_k^{v_{p_k}(b)}|b$ and thus $a|b$ as desired.